NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

DIAMETER TOLERANCES

NOMINAL OD	OD TOLERANCE
UP TO 3/32 EXCL	±.001
3/32 TO 3/16 EXCL	±.0015
3/16 TO 1/2 EXCL	±.003
1/2 TO 1 EXCL	±.004
1 TO 1-1/2 EXCL	±.005
1-1/2 TO 2 EXCL	±.006
2 TO 2-1/2 EXCL	±.007
2-1/2 TO 3-1/2 EXCL	±.010
3-1/2 TO 5 INCL	±.015

OVALITY IS THE DIFFERENCE BETWEEN MAX AND MIN OUTSIDE DIAMETERS MEASURED AT ANY ONE CROSS SECTION. THERE IS NO ADDITIONAL TOLERANCE FOR OVALITY ON TUBES HAVING A NOMINAL WALL THICKNESS OF MORE THAN 3 PERCENT OF THE OD.

AN ADDITIONAL OVALITY ALLOWANCE, OF TWICE THE TUBULAR OD TOLERANCE SPREADS SHOWN ABOVE, APPLIED ONE-HALF PLUS AND ONE-HALF MINUS, IS APPLIED TO TUBES HAVING A NOMINAL WALL THICKNESS OF 3 PERCENT OR LESS OF THE NOMINAL OD. THE AVERAGE OF THE MAX AND MIN OD READINGS SHOULD FALL WITHIN THE OD TOLERANCES AS SHOWN IN THE TABLE ABOVE.

EXCLUSIVE AS USED HEREIN IS APPLICABLE ONLY TO THE LARGER DIAMETER LIMIT.

WALL THICKNESS TOLERANCES

NOMINAL WALL THICKNESS	OUTSIDE DIAMETER			
	3/16 TO 1 EXCL	1 TO 2 EXCL	2 TO 4 EXCL	4 TO 5 INCL
TO .025, INCL	±.002	±.003	----	----
OVER .025 THRU .028, INCL	±.003		±.004	
OVER .028 THRU .032, INCL		±.004	±.005	±.005
OVER .032 THRU .035, INCL				
OVER .035 THRU .049, INCL				±.006
OVER .049 THRU .058, INCL	±.004	±.005	±.006	±.007
OVER .058 THRU .065, INCL	±.005			
OVER .065 THRU .072, INCL				
OVER .072 THRU .083, INCL				
OVER .083 THRU .095, INCL				
OVER .095 THRU .109, INCL		±.006	±.007	
OVER .109 THRU .120, INCL				
OVER .120 THRU .134, INCL	----	----	±.008	±.008
OVER .134 THRU .149, INCL	----	----		
OVER .149 THRU .165, INCL	----	----		

EXCLUSIVE AS USED HEREIN APPLIES ONLY TO THE LARGER DIAMETER LIMIT.

FOR REDRAWN WELDED STAINLESS, WALL TOLERANCES ARE ±10 PERCENT.

LENGTH TOLERANCES

NOMINAL LENGTH, LINEAR FEET	OUTSIDE DIAMETER	TOLERANCE
TO 4 INCL	TO 2 INCL	+1/16, -0
	OVER 2 TO 4 INCL	+3/32, -0
	OVER 4	+1/8, -0
OVER 4 TO 10 INCL	TO 2 INCL	+3/32, -0
	OVER 2	+1/8, -0
OVER 10 TO 24 INCL	ALL	+3/16, -0

FOR EVERY 10 FT OR FRACTION THEREOF OVER AND ABOVE 24 FT THERE SHOULD BE AN ADDITIONAL ALLOWANCE OF +1/16, -0.

STRAIGHTNESS: TOLERANCE: STANDARD TOLERANCE FOR STRAIGHTNESS IS .030 INCHES IN A LENGTH OF 3 FT FOR TUBING LENGTHS 24 FT AND UNDER.

INSIDE WELDING FLASH

WALL THICKNESS	MAXIMUM PERMISSIBLE HEIGHT OF WELDING FLASH
.049 AND UNDER	.004
OVER .049 THRU .065, INCL	.008
OVER .065	.012

DIMENSIONS IN INCHES.

(A) DRAWING REVISED.

THIS IS A DESIGN STANDARD. NOT TO BE USED AS A PART NUMBER.

THIS DOCUMENT HAS BEEN PROMULGATED BY THE DEPARTMENT OF DEFENSE AS THE MILITARY STANDARD TO LIMIT THE SELECTION OF THE ITEM, PRODUCT, OR DESIGN COVERED HEREIN IN ENGINEERING, DESIGN, AND PROCUREMENT. THIS STANDARD SHALL BECOME EFFECTIVE NOT LATER THAN 90 DAYS AFTER THE LATEST DATE OF APPROVAL SHOWN.

APPROVED 26 Jan 53 REVISED (A) 23 Aug 57

CUSTODIANS	OTHER INT.	MILITARY STANDARD	MS33531 (ASG)
Navy - BuAer Air Force	A - N - AF -	TOLERANCE, WELDED CORROSION-RESISTANT STEEL TUBING	
PROCUREMENT SPECIFICATION NONE		SUPERSEDES:	SHEET 1 OF 1

☆ U. S. GOVERNMENT PRINTING OFFICE: 1971 -433-696/4782

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T17:04Z
Check the source to verify that this is the current version before use.